Hooke’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect spring displacement with restoring force.

SUCCESS CRITERIA - I can...

1. Measure displacement from the natural length.
2. Use spring constant in N/m.
3. Point the spring force back toward equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “spring constant” and “displacement” to explain your model or result. Sentence starters are optional.

spring constant: how much an ideal spring’s force changes per stretch

displacement: change from initial position to final position, with direction

equilibrium: zero net force and zero acceleration

01 Warm-up: bring back an earlier idea

A wheel rolls down a ramp, losing 15 J gravitational energy. Its center-of-mass kinetic energy gain is 12 J with negligible losses. What energy term is missing?

Word help: negligible = small enough to leave out of this model

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_{spring} = -kx$; $|F_{spring}| = k|x|$.
x is displacement from natural length; k is spring constant. The force restores the spring, within its linear range.

Worked reasoning: A spring with k=100 N/m stretches 0.03 m, giving force magnitude kx=3 N. Its force on the attached object points toward equilibrium.

Your turn: Sketch equilibrium and stretched positions, then draw the restoring-force arrow.

Sentence starter: The … in my model represents … . Its label shows … .



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A spring is stretched right.
Draw the force it exerts on the attached block. What changes if compressed left?

Sentence starter: The … in my model represents … . Its label shows … .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: 1 2 3

04 Apply the physics

You test an ideal spring with k = 100 N/m.
Calculate restoring-force magnitudes for stretches of 0.03 m and 0.06 m.
Label both displacements and show the force direction relative to equilibrium.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says k doubles when the stretch doubles.
Explain which quantity actually doubles for the same ideal spring.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 5 N force acts forward through 7 m.
Find work and show the force/displacement directions.
Label force and displacement, including their relative direction.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

k=200 N/m and compression=0.04 m.
Find force size and describe direction.
Mark the displacement from equilibrium and draw the restoring-force direction.

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two springs have k=50 and 100 N/m.
Find stretches that both give 4 N, and compare their stiffness.
Mark the displacement from equilibrium and draw the restoring-force direction.

Sentence starter: Both … . They differ in … because … .



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3